AbdElrahman Mohammed

Software Engineering Student (Class of 2027) | AI & Back-End Developer | 2nd Place, AI Hackathon (Medical RAG)

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SUMMARY

Final-year Software Engineering student (graduating May 2027) who builds working AI prototypes end to end. Won **2nd place at the AI Hackathon (Aug 2026)** with Care for You, a medical RAG assistant in Python/FastAPI combining hybrid retrieval, LLM APIs, citation validation, and personalized memory. Builds production-style C#/ASP.NET Core back ends with Angular front ends, applying Clean Architecture, DDD, and CQRS from a Link Development internship, and takes business problems from requirements to tested, demo-ready products.

EDUCATION

B.Sc. in Software Engineering — Al-Azhar University, Cairo, Egypt Aug 2022 – May 2027 (expected)

AI HACKATHON PROJECT

Care for You — Medical RAG Assistant | 2nd Place, AI Hackathon Aug 2026

Creativa Innovation Hubs × Orange Digital Center Egypt × Instant Software Solutions · 17 governorates, 15,000+ applicants

Team project · Python, FastAPI, Qdrant, LLM APIs, Supabase, AWS · github.com/mohamed-hossam1/hackathon-rag

- **Problem & solution:** with my team, built a clinical decision-support assistant that answers medical questions from official guidelines (e.g., WHO) with cited sources and abstains when evidence is weak. I worked end to end across ingestion, retrieval, generation, and the API, and pitched it in the final round.
- **Data handling:** ingestion of PDF, DOCX, DOC, Markdown, and TXT with OCR fallback for scanned pages, table extraction, text cleaning, semantic and recursive chunking, and deduplication in a background queue.
- **Retrieval:** hybrid search combining Qdrant vector search (BGE embeddings) with BM25 keyword search, fused with Reciprocal Rank Fusion and reranked by a BGE cross-encoder.
- **Generative AI:** grounded answers via OpenAI-compatible LLM APIs (Groq/OpenRouter) with streaming responses; an LLM-based citation validator scores evidence and blocks unsupported answers.
- **Personalization:** an LLM detects personal medical facts (conditions, allergies, medications) in Arabic or English messages and, with user consent, saves them as memory to tailor later answers; Supabase auth and per-user chat history.
- **Evaluation & deployment:** compared retrieval strategies with Precision@3/@5 and an LLM-as-a-judge; deployed the API to AWS EC2 via GitHub Actions, with S3 document storage.

SKILLS

GenAI & NLP APIs: OpenAI SDK (Groq, OpenRouter endpoints), Hugging Face models (BGE embeddings and reranker), RAG, prompt design, LLM citation validation, LLM-as-a-judge

Machine Learning & AI: embedding-based semantic search, cross-encoder relevance ranking, Reciprocal Rank Fusion, LLM-based text classification, retrieval evaluation (Precision@K)

Data Handling (cleaning, structuring, storing): document parsing (PyMuPDF, python-docx), text cleaning, table extraction, chunking, deduplication; SQL Server, PostgreSQL (Supabase), MySQL, Qdrant

Python (backend + AI logic): FastAPI, Pydantic, Flask, background task queues, streaming APIs

Frontend / Backend Development: Angular, TypeScript, HTML, CSS, Bootstrap; ASP.NET Core Web API (C#), EF Core, REST, OAuth 2.0 / OpenID Connect, JWT

Pitching & Storytelling: pitched Care for You in the AI Hackathon final (2nd place); volunteer technical instructor

Computer Vision: OCR for scanned and image-only documents (Tesseract, Pillow)

Advanced Personalization: user medical memory (conditions, allergies, medications) and chat history used to tailor answers

Chatbots (English/Arabic): medical chatbot with streaming and multi-chat history; LLM processing of Arabic and English messages

Architecture, Cloud & Tools: Clean Architecture, DDD, CQRS, xUnit, Testcontainers, AWS (EC2, S3), GitHub Actions, Git, Swagger; AI-assisted development (GitHub Spec Kit, Claude Code, Codex)

Programming Languages: Python, C#, TypeScript, SQL, Java, C, PHP

EXPERIENCE

Back-End .NET Intern — Link Development Jul 2026 – Aug 2026

Talabat-like food-delivery platform · .NET, EF Core, SQL Server, Duende IdentityServer, Angular · Talabat_API · talabat-frontend

- Applied CQRS, Domain-Driven Design, system design, and software architecture from the training by building a Talabat-like food-delivery platform, which I continued developing after the internship.
- Implemented a layered Clean Architecture (Domain, Application, Infrastructure) with separate Customer, Delivery, and Admin APIs and an identity server (OAuth 2.0 / OpenID Connect, authorization code with PKCE, role-based access).
- Modeled DDD aggregates and domain services for cart expiry, checkout validation, and delivery assignment, using CQRS-style use-case handlers and optimistic concurrency (RowVersion) to prevent conflicting courier assignments.
- Built the Angular customer app end to end (browse, menu, cart, checkout, order history) against the APIs' OpenAPI contracts; covered the back end with 600+ automated tests, including architecture tests and SQL Server integration tests using Testcontainers.

Technical Support Specialist — Arkan for Digital Technology May 2025 – May 2026

- Supported business clients using ERP, POS, and invoicing software, resolving day-to-day issues in their sales and invoicing operations.
- Guided clients through an e-invoicing application that automates invoice submission to the Egyptian Tax Authority (ETA) portal.

SAP Basis Intern — IT Department, Amer Group Jul 2022 – Aug 2022

PROJECTS

HR Management System (independent project) | *ASP.NET Core 8, Angular, SQL Server, EF Core, ASP.NET Identity, xUnit*
github.com/engakhattab/DEPI_Angular_project

- Independently built a full-stack HR platform covering employees, departments, vacation and transportation requests, attendance, compensation and salary history, employee documents, dashboards, and audit logs.
- Turned HR policy into enforced business rules: leave balance and overlap checks, a three-working-day notice period, approval workflows, and an employee lifecycle in which termination immediately revokes access.
- Implemented policy-based access for System Admin, HR Admin, Manager, and Employee roles with scoped data: managers see only their direct reports, and compensation stays hidden from employees.
- Structured the solution into Domain, Application, Infrastructure, and API layers with EF Core migrations and Swagger documentation, verified by 390+ automated xUnit tests.

IP Blocker API | *ASP.NET Core, REST, external data API*
github.com/engakhattab/IpBlocker

- Built a Web API that blocks requests by country (permanent and time-limited), integrating the IPGeolocation.io API behind a service interface, with a background cleanup worker, request logging, and unit tests.

TRAINING & CERTIFICATIONS

Front-End Development with Angular — Digital Egypt Pioneers Initiative (DEPI)	Jun 2025 – Dec 2025
Full-Stack .NET Web Development — Digital Egypt Pioneers Initiative (DEPI)	Apr 2024 – Nov 2024
ASP.NET — Information Technology Institute (ITI)	Jul 2024 – Sep 2024
Software Engineering Program, Back-End Specialization (Python, C) — ALX Africa	Nov 2023 – Nov 2024
45-Hour Training Course and Final Project for a Potential Startup — Innove Egypt	Jan 2024
Web Development Challenger Track — Ministry of Communications and Information Technology	Sep 2022
Full Stack Development Track (1 Million Arab Coders Initiative) — Udacity	Jul 2020

LEADERSHIP & VOLUNTEERING

Back-End Instructor (.NET), Volunteer — Az-SENCS Oct 2024 – May 2025

- Delivered technical presentations and training sessions on back-end development with .NET, and mentored team members on coding and problem-solving.

LANGUAGES

Arabic (Native) | English (B2)